



Methodology

WALMART SALES FORECASTING

A detailed walkthrough of the data preprocessing, modeling, and evaluation methodology used to forecast weekly sales across 45 stores.

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Repository: github.com/sidducv0528/Walmart-Sales-Forecasting

1. Data Preprocessing

1 Date parsing

The `Date` column is parsed from its original `DD-MM-YYYY` string format into a proper datetime type, which is required before any time-series operations (resampling, frequency setting, seasonal modeling) can be applied.

2 Sorting

Data is sorted by `Store`, then `Date`, to guarantee chronological order within each store's time series — SARIMA is order-sensitive and will silently produce wrong results on unsorted data.

3 Frequency alignment

Each store's series is set to a weekly-Friday frequency (`W-FRI`) using `asfreq()`. This confirms there are no gaps in the weekly cadence and gives `statsmodels` the regular time index it needs to fit a seasonal model correctly.

4 Per-store split

Rather than pooling all 45 stores into one series, the data is grouped by `Store` and each store's history is treated as an independent time series — both for modeling and evaluation. This reflects the EDA finding that stores differ meaningfully in scale and behavior.

Verification: all 45 stores were confirmed to have exactly 143 weekly rows each, spanning Feb 2010 – Oct 2012, with no missing weeks after alignment.

2. EDA Methodology

2.1 Correlation Analysis

Pearson correlation was computed between `Weekly_Sales` and each candidate driver — `Unemployment`, `CPI`, and `Temperature` — at two levels:

- **Aggregate level** — correlation across the full dataset, to gauge overall relationship strength.
- **Per-store level** — correlation computed independently within each store's data, to check whether the aggregate result masks store-specific sensitivity.

This two-level approach matters: a near-zero aggregate correlation could either mean "no store cares about this factor" or "different stores respond in opposite directions and cancel out." The per-store breakdown distinguishes between these, and is what surfaced the finding that a small subset of stores are meaningfully more unemployment-sensitive than the rest, even though the aggregate correlation is weak.

2.2 Seasonality Analysis

Average weekly sales were grouped by calendar month across all stores and all years, to visualize the shape of the yearly cycle independent of any single year's anomalies. Separately, mean sales were compared between `Holiday_Flag == 1` and `Holiday_Flag == 0` weeks to quantify the holiday sales lift as a single percentage figure.

2.3 Store Comparison

Stores were ranked by total sales across the full period. The top and bottom stores were compared directly (total sales, ratio between them) to quantify how much store-to-store variation exists — this became the basis for choosing a per-store modeling approach over a single pooled model.

3. Model Specification

3.1 Why SARIMA

The sales series shows two clear structural properties that point directly to a seasonal ARIMA family model:

- **Trend/non-stationarity** — handled by the non-seasonal differencing term `d=1`.
- **Strong yearly seasonality** — a 52-week cycle driven by the holiday season, handled by the seasonal order with `period=52`.

Simpler alternatives (plain linear regression on time, or non-seasonal ARIMA) were ruled out early: the EDA's seasonality analysis makes it clear that any model ignoring the 52-week cycle would systematically under-forecast every November–December.

3.2 Parameter Choice

SARIMA(1,1,1) (1,1,1,52)

Term	Value	Meaning
p	1	One autoregressive lag (non-seasonal)
d	1	One order of differencing to remove trend
q	1	One moving-average lag (non-seasonal)
P	1	One seasonal autoregressive lag
D	1	One order of seasonal differencing
Q	1	One seasonal moving-average lag
s	52	Seasonal period — one year of weekly data

This `(1,1,1) (1,1,1,52)` order was applied uniformly across all 45 stores rather than tuned individually. This is a known simplification and is flagged explicitly in Section 6 (Limitations) — per-store tuning (e.g. via `auto_arima`) is listed as a concrete next step, since different stores may have subtly different optimal orders.

3.3 Fitting Approach

Two separate model fits are performed per store:

1. **Evaluation fit** — trained on all data except the last 12 weeks, used purely to generate held-out predictions for scoring against known actuals.
2. **Production fit** — trained on the store's *entire* history, used to generate the genuine 12-week-ahead forecast into the future (where no ground truth exists yet).

This separation matters: using the evaluation model's forecast as the "real" forecast would mean the deployed forecast is missing the most recent 12 weeks of training data for no reason. Fitting a second, fuller model avoids that waste.

4. SARIMA vs. SARIMAX — An Honest Note

The dataset includes three columns that could serve as exogenous regressors in a true SARIMAX model: `Holiday_Flag`, `CPI`, and `Unemployment`. This section documents the actual evaluation behind the decision not to use them, rather than treating the choice as a default.

4.1 The Test

```
exog_check = df.groupby('Store').apply(
    lambda x: pd.Series({
        'corr_unemployment': x['Weekly_Sales'].corr(x['Unemployment']),
        'corr_cpi': x['Weekly_Sales'].corr(x['CPI']),
    })
)

print(f"Unemployment: {exog_check['corr_unemployment'].abs().mean():.4f}")
print(f"CPI: {exog_check['corr_cpi'].abs().mean():.4f}")

holiday_lift = (df[df.Holiday_Flag == 1]['Weekly_Sales'].mean() /
                df[df.Holiday_Flag == 0]['Weekly_Sales'].mean() - 1) * 100
```

4.2 The Reasoning

1. **Holiday_Flag** — the seasonal term `(..., 52)` already models the yearly cycle that produces the holiday spike. Adding `Holiday_Flag` as a separate 0/1 exogenous input would be largely redundant with information the seasonal component already captures.
2. **CPI and Unemployment** — both are macroeconomic indicators that change gradually month-to-month. Average per-store correlation with weekly sales is weak, and given how slowly these values move, they are unlikely to add meaningful signal to a short 12-week-ahead forecast.

4.3 What This Means for the Model Name

The model is implemented using `statsmodels.tsa.statespace.sarimax.SARIMAX`, which supports an `exog` parameter for exactly this purpose — but no exogenous array is passed in this project. Functionally, the fitted model is a **seasonal ARIMA (SARIMA)** model. The class name in the code is SARIMAX because that's the statsmodels class that implements both SARIMA and SARIMAX (SARIMAX is the superset; omitting `exog` reduces it to SARIMA). This is called out explicitly here, in the notebook, and in the project README, so it doesn't read as an accidental misnomer.

5. Evaluation Methodology

5.1 Train/Test Split

For each store, the final 12 weeks of the 143-week history are held out as a test set; the model trains only on the preceding weeks. This mirrors the actual deployment scenario — forecasting 12 weeks into the unknown future — rather than a random train/test split, which would leak future information backward through the seasonal terms.

5.2 Metrics

Metric	Formula	What it tells you
MAE	$\text{mean}(\text{actual} - \text{predicted})$	Average error, in the same units as sales (\$)
RMSE	$\text{sqrt}(\text{mean}((\text{actual} - \text{predicted})^2))$	Like MAE, but penalizes large individual misses more heavily
MAPE	$\text{mean}(\text{actual} - \text{predicted} / \text{actual}) \times 100$	Error as a percentage — comparable across stores of very different sizes

MAPE is used as the primary rating metric specifically *because* it's scale-independent — MAE and RMSE in raw dollars aren't directly comparable between a \$200K/week store and a \$2M/week store, but a 3% MAPE means the same thing for both.

5.3 Rating Thresholds

MAPE	Rating
< 10%	Excellent
< 20%	Good
≥ 20%	Needs Improvement

These thresholds are a judgment call, not a statistical standard — chosen to be interpretable at a glance in the dashboard, and reasonably aligned with common informal benchmarks for retail demand forecasting.

6. Limitations & Next Steps

- **Fixed order across all stores.** The same `(1,1,1)(1,1,1,52)` order is applied to every store. Per-store tuning (grid search or `auto_arima`) would likely improve accuracy for stores whose dynamics differ from the average — Store 35, the one clear underperformer at 14.15% MAPE, is a natural first candidate to investigate.
- **No true exogenous variables tested empirically.** The decision to exclude CPI/Unemployment/Holiday_Flag was based on correlation analysis, not on a direct MAPE comparison between a SARIMA and a SARIMAX fit. Running that comparison directly would either confirm or overturn the current assumption.
- **Static evaluation window.** A single 12-week holdout is used per store, rather than rolling-origin cross-validation, which would give a more robust estimate of forecast accuracy across different points in time.